MIL-M-48146 (MU)
8 March 1973
SUPERSESSION
(see 6.5)

MILITARY SPECIFICATION

MOLYBDENUM TRIOXIDE
(FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers one grade of Molybdenum Trioxide intended for use in ammunition (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Shipping, Paper
PPP-D-723- Drums, Fiber
PPP-D-729- Drums, Metal, 55 Gallon (for shipment of noncorrosive material)

MILITARY

MIL-D-112 - Packaging and Packing for Overseas Shipment Drums, Plywood (for drums whose weight of contents does not exceed 200 pounds.)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1233- Procedures for Determining Particle Size, Particle Size Determination and Packing Density of Powder Material
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATION

ASTM Designation E300-70 - Recommended Practice for Sampling Industrial Chemicals

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Ill. 60606).

AMERICAN TRUCKING ASSOCIATION PUBLICATION

National Motor Freight Classification Rules and Container Specifications. (Application for copies should be addressed to the American Trucking Association, 1424 16th Street N.W., Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 The requirements for molybdenum trioxide shall comply with Table I when determined as specified in the applicable paragraphs of section 4.4.

TABLE I

Property	Percent	Test Paragraph
Assay, minimum (min.)	99.5	4.4.1
Insoluble in ammonium hydroxide, maximum (max.)	0.010	4.4.2
Chloride (Cl), max.	0.002	4.4.3
Nitrate (NO_3), max.	0.003	4.4.4
Arsenate, phosphate and silicate (as SiO_2), max.	0.001	4.4.5

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Sulfate (SO_4), max.	0.020	4.4.6
Ammonium (NH_4), max.	0.002	4.4.7
Heavy metals (as Pb), max.	0.005	4.4.8

3.2 Average particle size.-The average particle size of molybdenum trioxide shall be 8 plus or minus 4 microns when determined as specified in paragraph 4.4.9.

3.3 Workmanship.-The material furnished under this specification shall be free flowing and free of lumps, dirt, chips and other foreign material.

3.4 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define the terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all requirements and quality assurance provisions specified in this specification.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications, shall be made available upon request by the contracting officer.

d. Quantity of product in the lot.

e. Date submitted.

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The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to initiation of regular production the contractor shall submit a first article sample consisting of 150 grams of molybdenum trioxide in accordance with instructions issued by the Contracting Officer for evaluation in accordance with paragraph 4.2.2. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. (see 6.1)

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation.-A lot shall consist of one or more batches of molybdenum trioxide produced by one manufacturer in accordance with the same specification or same specification

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revision under one continuous set of operating conditions. Each batch shall consist of that quantity of molybdenum trioxide that has been subject to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105). Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each Minor Defect.

4.3.2.1 Drum, prior to filling (Polyethylene liner)

Categories	Defects	Method of Inspection	Code No. (see 6.4)
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Liner cut, torn or punctured...	Visual	01001
Minor: None defined.			

4.3.2.1.1 Drum, prior to filling (alternative method)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Laminations or coatings incomplete.....	Visual	01002
Minor: None defined.			

4.3.2.2 Drum, prior to closing (Polyethylene liner)

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Liner not completely sealed....	Visual/ Manual	02001
Minor:	None defined.		

4.3.2.3 Drum, sealed (steel)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Closing ring damaged or improperly positioned (drum cannot be closed properly).....	Visual	03001
Minor:	AQL 0.65 percent		
201.	Marking misleading, missing or unidentifiable.....	Visual	03002

4.3.2.4 Drum, sealed (fiber)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of contents.....	Balance(CD)	04001
102.	Closure incomplete or damaged to the extent that contents sift out.....	Visual	04002
Minor:	0.65 percent		
201.	Marking misleading, missing or unidentifiable.....	Visual	04003

4.3.2.5 Drum, sealed (plywood)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.40 percent		
101.	Weight of contents.....	Balance(CD)	05001
102.	Strapping missing, broken or loose.....	Visual/ Manual	05002
Minor:	AQL 1.0 percent		
201.	Strapping misplaced.....	Visual	05003
202.	Marking misleading, missing or unidentifiable.....	Visual	05004

4.3.3 Sampling for test 4.4.1 through 4.4.9.-A representative sample of approximately 150 grams of Molybdenum Trioxide shall be selected from each batch using ASTM Procedure E300-70 for solids. The inspection of the samples shall be in accordance with MIL-STD-1235, CSP-1 Plan, Inspection Level II, AQL 6.5%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.4. If any of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The classification and code number shall be as given in Table II.

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TABLE II

Assay	Major Defect	Code No. 06001
Insoluble in ammonium hydroxide	Major Defect	Code No. 07001
Chloride (Cl)	Major Defect	Code No. 08001
Nitrate (NO_3)	Major Defect	Code No. 09001
Arsenate, phosphate and silicate (as SiO_2)	Major Defect	Code No. 10001
Sulfate (SO_4)	Major Defect	Code No. 11001
Ammonium (NH_4)	Major Defect	Code No. 12001
Heavy metals (as Pb)	Major Defect	Code No. 13001
Average particle size	Major Defect	Code No. 14001

4.3.4 Inspection equipment.-The inspection equipment required to perform the examination and tests prescribed in this section is identified under the "Method of Inspection" heading in 4.3.2 (see 6.3).

4.4 Test Methods and Procedures

4.4.1 Assay

4.4.1.1 Lead acetate reagent (10 percent) solution.- Dissolve 100 grams of lead acetate [$(CH_3COO)_2$ Pb. $3H_2O$] in distilled water. If necessary, add a few drops of acetic acid to clear the solution and dilute to 1000 ml.

4.4.1.2 Methyl orange indicator.-Dissolve 0.10 grams of methyl orange in 100 ml of distilled water.

4.4.1.3 Procedure.-Approximately 1 gram of sample, accurately weighed, shall be dissolved in 10 ml of distilled water plus 1 ml of ammonium hydroxide. The solution shall be transferred to a 250 ml volumetric flask diluted to volume and mixed throughly. A 50.0 ml aliquot shall be transferred to a 600 ml beaker to which the following shall be added: 250 ml of distilled water, 20 grams of ammonium chloride, 15 ml of hydrochloric acid and 0.15 ml

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of methyl orange indicator solution (see 4.4.1.2). The solution shall be heated nearly to boiling and 18 ml of the 10 percent lead acetate reagent solution shall be added (see 4.4.1.1). To the hot and constantly stirred solution add slowly a saturated solution of ammonium acetate until the solution is alkaline, and then an excess of 15 ml of the acetate solution. Digest on the hot plate in the covered beaker at a temperature just below boiling until the precipitate has settled (from one half to one hour). Filter the mixture through a Gooch crucible which has been ignited and weighed. Wash the crucible eight times with a solution containing 10 ml of nitric acid and 100 ml of a saturated solution of ammonium acetate which has been diluted to 1000 ml with distilled water, and then wash three times with hot water. Ignite the crucible in a muffle furnace at 560°C to 625°C until constant weight is obtained. The crucible and contents shall be cooled in a desiccator and weighed as lead molybdate ($PbMoO_4$). Calculate the percentage of molybdenum trioxide as follows:

$$\text{Percent Molybdenum Trioxide} = \frac{0.3921 \times W_1 \times 100}{W_2}$$

Where: W_1 = weight of residue, in g
W_2 = weight of sample in aliquot used, in g

4.4.2 Insoluble in dilute ammonium hydroxide.- Approximately 10 grams of sample, accurately weighed, shall be dissolved in 120 ml of dilute ammonium hydroxide (1 plus 6) (see 6.8) and then heated to boiling and digested in a covered beaker on a steam bath for 2 hours. Filter through a tared medium porosity filtering crucible, wash thoroughly with distilled water and dry at 105°C to constant weight. The residue shall be calculated as percent insoluble in dilute ammonium hydroxide as follows:

$$\text{Percent insoluble in dilute ammonium hydroxide} = \frac{100A}{W}$$

Where: A = Weight of residue, in g
W = Weight of sample used, in g

4.4.3 Chloride

4.4.3.1 Silver nitrate reagent solution.-Dissolve 17 grams of silver nitrate ($AgNO_3$) in 1000 ml of distilled water.

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4.4.3.2 Chloride standard (0.01 mg Cl in 1 ml).-Dissolve 0.165 grams of sodium chloride (NaCl) in distilled water and dilute to 100 ml. Dilute 10 ml of this solution to 1000 ml.

4.4.3.3 Procedure.-Accurately weigh approximately 1 gram of sample and dissolve in 4 ml of ammonium hydroxide, heat on the steam bath until completely dissolved, and evaporate to dryness. Dissolve the residue in 20 ml of distilled water, filter if necessary through a chloride-free filter, add 4 ml of nitric acid, and dilute to 30 ml with distilled water. To 15 ml of the sample solution add 1 ml of silver nitrate reagent solution (see 4.4.3.1). Any turbidity should not exceed that produced by 1 ml of chloride standard (see 4.4.3.2) in an equal volume of solution containing the quantities of reagents used in the test. The comparison is best made by superimposing a tube containing the remaining 15 ml of sample solution and 1 ml of water over the tube containing the standard turbidity and placing a tube containing 16 ml of water below the tube containing the sample and added reagents. Both turbidities are thus viewed through the same depth and color of solution. The comparison tubes may be machine-made vials, long style, of about 20 ml capacity.

4.4.4 Nitrate

4.4.4.1 Indigo carmine reagent solution.-Dissolve 1 gram of FD and C Blue No. 2, (see 6.7), dried at 105°C, in a mixture of 800 ml of distilled water and 110 ml of sulfuric acid, and dilute to 1000 ml.

4.4.4.2 Procedure.-Approximately 1 gram of sample, accurately weighed, shall be triturated with 9 ml of distilled water and 1 ml of sodium chloride solution containing 5 mg of sodium chloride. Add 0.20 ml of indigo carmine reagent solution (see 4.4.4.1) and 10 ml of sulfuric acid. The blue color shall not be completely discharged in 5 minutes.

4.4.5 Arsenate, phosphate, and silicate

4.4.5.1 Silicate standard (0.01 mg SiO_2 in 1 ml).-Dissolve 0.473 grams of sodium silicate ($Na_2SiO_3.9H_2O$) in 100 ml of distilled water in a platinum or polyethylene dish. Dilute 10 ml of this solution (polyethylene graduate) to 1000 ml in a polyethylene bottle.

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4.4.5.2 Bromine water - A saturated aqueous solution of bromine. Add enough liquid bromine to distilled water in a bottle so that when the mixture is shaken, undissolved bromine remains in a separate phase.

4.4.5.3 Stannous Chloride in hydrochloric acid (2 percent solution).-Dissolve 2 grams of stannous chloride ($SnCl_2.2H_2O$) in hydrochloric acid and dilute to 100 ml with hydrochloric acid.

4.4.5.4 Procedure.-Accurately weigh approximately 1.25 grams of sample and dissolve in 60 ml of distilled water and enough silica-free ammonium hydroxide to effect dissolution (about 4 ml) in a platinum dish. For the control dissolve approximately 0.25 grams of sample, accurately weighed, in 60 ml of distilled water and enough silica-free ammonium hydroxide to effect dissolution and add 1 ml of silicate standard (see 4.4.5.1). Heat on the steam bath until the solutions are neutral as determined with an external indicator (at least 30 minutes). Cool and adjust the pH to 3 to 4 with an external indicator. Transfer to beakers and dilute to 80 ml with distilled water. Add enough bromine water (see 4.4.5.2) to impart a distinct yellow color to the solutions. Adjust the pH of each solution to 1.8 with dilute hydrochloric acid (1 + 9) (using a pH meter). Heat just to boiling and allow to cool to room temperature. (If a precipitate forms, it will dissolve when the solution is acidified in the next operation). Add 10 ml of hydrochloric acid and dilute to 100 ml. Transfer the solutions to separatory funnels and add 30 ml of 4-methyl-2-pentanone (ketone) and 1 ml of butyl alcohol. Shake vigorously and allow to separate. Draw off and discard the aqueous phase. Wash the ketone phase 3 times with 10 ml portions of dilute hydrochloric acid (1 + 99), discarding the aqueous phase each time. To the washed ketone phase add 10 ml of dilute hydrochloric acid (1 + 99) to which has just been added 0.2 ml of a freshly prepared 2 per cent solution of stannous chloride in hydrochloric acid (see 4.4.5.3) Any blue color in the solution of the sample should not exceed that in the control.

4.4.6 Sulfate

4.4.6.1 Barium chloride reagent solution.-Dissolve 120 grams of barium chloride ($BaCl_2.2H_2O$) in distilled water and dilute to 1000 ml.

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4.4.6.2 Sulfate standard (0.01 mg SO_4 in 1 ml).-Dissolve 0.148 grams of anhydrous sodium sulfate (Na_2SO_4) in distilled water and dilute to 100 ml. Dilute 10 ml of this solution to 1000 ml.

4.4.6.3 Procedure.-Boil approximately 1 gram of sample, accurately weighed, with 15 ml of dilute nitric acid (1 + 2) for 5 minutes. Cool thoroughly, dilute to 50 ml with distilled water, mix well, and filter through a medium filter paper. Evaporate 10 ml of the filtrate to dryness on the steam bath. Dissolve the residue in 4 ml of distilled water plus 1 ml of dilute hydrochloric acid (1 + 19), filter if necessary through a small medium filter paper, wash with two 2-ml portions of distilled water, and dilute to 10 ml. Add 1 ml of barium chloride reagent solution (see 4.4.6.1). Any turbidity should not exceed that produced by 4 ml of sulfate (SO_4) standard (see 4.4.6.2) in an equal volume of solution containing the quantities of reagents used in the test. Compare 10 minutes after adding the barium chloride to the sample and standard solutions.

4.4.7 Ammonium

4.4.7.1 Sodium hydroxide reagent solution.-Dissolve 100 grams of sodium hydroxide (NaOH) in distilled water and dilute to 1000 ml.

4.4.7.2 Nessler Reagent.-Dissolve 143 grams of sodium hydroxide in 700 ml of distilled water. Dissolve 50 grams of red mercuric iodide and 40 grams of potassium iodide in 200 ml of distilled water. Pour the iodide solution into the hydroxide solution and dilute to 1000 ml. Allow to settle and use the clear supernatant liquid. Nessler reagent prepared by any of the recognized methods may be used, but the reagent described reacts promptly and consistently.

4.4.7.3 Ammonium standard (0.01 mg NH_4 in 1 ml).-Dissolve 0.296 grams of ammonium chloride (NH_4Cl) in distilled water and dilute to 100 ml. Dilute 10 ml of this solution to 1000 ml.

4.4.7.4 Procedure.-Dissolve approximately 1 gram of sample, accurately weighed, in 15 ml of freshly boiled 10 percent sodium hydroxide reagent solution (see 4.4.7.1) and dilute to 100 ml with distilled water. To 50 ml of the solution add 2 ml of Nessler reagent (see 4.4.7.2). Any color should not exceed that produced by 1 ml of ammonium standard (see 4.4.7.3) in an equal volume of solution containing 7.5 ml of 10 percent sodium hydroxide reagent solution and 2 ml of Nessler reagent.

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4.4.8 Heavy Metals (as Pb)

4.4.8.1 Lead Stock Solution (0.10 mg Pb in 1 ml).-Dissolve 0.160 grams of lead nitrate [$(Pb(NO_3)_2$] in 100 ml of dilute nitric acid (1 + 99) and dilute to 1000 ml. The solution should be prepared and stored in containers free from lead. Its strength should be checked every few months to determine whether the lead content has changed by reaction with the container.

4.4.8.2 Lead Standard for Heavy Metals Test (0.01 mg Pb in 1 ml).-Dilute 10 ml of the lead stock solution to 100 ml with distilled water. This dilute solution must be freshly prepared.

4.4.8.3 Hydrogen Sulfide Water.-Saturate distilled water with hydrogen sulfied gas. This solution must be freshly prepared.

4.4.8.4 Procedure.-Dissolve approximately 1 gram of sample, accurately weighed, in 15 ml of 10 percent sodium hydroxide solution, add 2 ml of ammonium hydroxide, and dilute to 40 ml with distilled water. For the control add 2.5 ml of lead standard (see 4.4.8.2) to a 10 ml aliquot of the solution and dilute to 40 ml with distilled water. For the sample dilute the remaining solution to 40 ml with distilled water. Add 10 ml of freshly prepared hydrogen sulfide water to each and mix (see 4.4.8.3). Any color in the solution of the sample should not exceed that in the control.

4.4.9 Average particle size.-The average particle size shall be determined in accordance with Method 100 of the latest revision of MIL-STD-1233.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level A

5.1.1 Drums.-Unless otherwise specified in the contract or purchase order, molybdenum trioxide shall be packed in a fiber, plywood or metal drum in accordance with Specifications PPP-D-723 (Type III, Grade A), MIL-D-112, and PPP-D-729 (Type III, Full Removable Head Type), respectively. Fiber and plywood drums shall be furnished with a nominal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure.

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(Alternately, an equivalent degree of protection may be furnished by suitable laminations or coatings of the inner surface of the drum). The drums shall be closed in accordance with Specification PPP-D-723, PPP-D-729 and MIL-D-112 as applicable.

5.1.1.2 Alternate method.-Molybdenum trioxide may be packed in paper shipping sacks in accordance with Table I, Construction No. 2X of Specification UU-S-48. Closure of sacks shall be in accordance with above Specification.

5.1.2 Level C.-Molybdenum trioxide shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipments or National Motor Freight Rules and Container Specifications for truck shipments, as applicable.

5.2 Marking.-In addition to any special marking required by the contract or purchase order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this document.
b. Description sheets shall be prepared for each lot in accordance with MIL-STD-1171.
c. Provisions for submission of first article samples.

6.2 Intended use.-The molybdenum trioxide is intended for use in igniter compositions.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

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6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detail drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria herein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture (s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c) Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commander, Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SARPA-QA-A-P, in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

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6.4 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.5 Supersession Data.-This specification includes the requirements of Purchase Description PA-PD-2571, dated 15 July 1964.

6.6 The tables of random numbers in MIL-HDBK-53 (Handbook 53) "Guide for Sampling Inspection" and ICC "Table of 105,000 Random Decimal Digits" have been found suitable.

6.7 FD&C Blue No 2, the disodium salt of 5,5-indigotindisulfonic acid ($C_{16}H_8N_2O_2(SO_3NA)_2$ is one of the dyes which may be certified under the Federal Food, Drug, and Cosmetic Act as suitable for use in foods, drugs, and cosmetics.

6.8 Dilutions referenced in Section 4.4 such as (1 + 6), (1 + 9), etc. are interpreted as 1 part of the specified reagent plus 6, 9, etc. parts of distilled water, as applicable.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 6810-A963

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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SUBMITTED BY *(Printed or typed name and address - Optional)*

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